

## # STRING AND TEXT MANIPULATION – INTERVIEW QUESTIONS

### # 1. Remove Spaces from Given Text

# Input: "he llo wor ld"

# Output: "helloworld"

```
def remove_spaces(text):  
    return "".join(text.split())
```

```
print(remove_spaces("he llo wor ld"))
```

# Problem: Write a function to remove all spaces from the input string.

# Explanation: Remove any whitespace characters.

```
def remove_all_spaces(text):  
    return "".join(text.split())
```

```
print(remove_all_spaces("he llo wor ld"))
```

### # 2. Reverse a String

# Input: "hello"

# Output: "olleh"

```
def reverse_string(text):  
    return text[::-1]
```

```
print(reverse_string("hello"))
```

# Problem: Write a function to reverse the characters in a string.

# Input: "hello"

# Output: "olleh"

```
def reverse_characters(text):
```

```
    return text[::-1]
```

```
print(reverse_characters("hello"))
```

# 3. Reverse a String After Removing Spaces

# Input: "he llo world"

# Output: "dlrowolleh"

```
def reverse_after_removing_spaces(text):
```

```
    text = "".join(text.split())
```

```
    return text[::-1]
```

```
print(reverse_after_removing_spaces("he llo world"))
```

# Problem: Write a function to reverse a string after removing all spaces.

# Input: "he llo world"

# Output: "dlrowolleh"

```
def remove_spaces_and_reverse(text):  
    text = "".join(text.split())  
    return text[::-1]  
  
print(remove_spaces_and_reverse("he llo world"))
```

# 4. Convert Snake Case to Camel Case

# Input: "my\_variable\_name"

# Output: "myVariableName"

```
def snake_to_camel(text):  
    words = text.split("_")  
    result = words[0]  
  
    for word in words[1:]:  
        result = result + word.capitalize()  
  
    return result  
  
print(snake_to_camel("my_variable_name"))
```

# Problem: Convert a string from snake\_case to camelCase.

# Input: "my\_variable\_name"

# Output: "myVariableName"

```
def convert_snake_case_to_camel_case(text):  
    words = text.split("_")  
    result = words[0]  
  
    for word in words[1:]:  
        result = result + word.capitalize()  
  
    return result  
  
print(convert_snake_case_to_camel_case("my_variable_name"))
```

#### # 5. Convert Snake Case to Pascal Case

# Input: "my\_variable\_name"

# Output: "MyVariableName"

```
def snake_to_pascal(text):  
    words = text.split("_")  
    result = ""  
  
    for word in words:  
        result = result + word.capitalize()  
  
    return result  
  
print(snake_to_pascal("my_variable_name"))
```

# Problem: Convert a string from snake\_case to PascalCase.

# Input: "my\_variable\_name"

# Output: "MyVariableName"

```
def convert_snake_case_to_pascal_case(text):  
    words = text.split("_")  
    result = ""  
  
    for word in words:  
        result = result + word.capitalize()  
  
    return result  
  
print(convert_snake_case_to_pascal_case("my_variable_name"))
```

# 6. Convert Camel Case to Snake Case

# Input: "myVariableName"

# Output: "my\_variable\_name"

```
def camel_to_snake(text):  
    result = ""  
  
    for char in text:  
        if char.isupper():  
            result = result + "_" + char.lower()  
        else:  
            result = result + char
```

```
    return result
```

```
print(camel_to_snake("myVariableName"))
```

```
# Problem: Convert a string from camelCase to snake_case.
```

```
# Input: "myVariableName"
```

```
# Output: "my_variable_name"
```

```
def convert_camel_case_to_snake_case(text):
```

```
    result = ""
```

```
    for char in text:
```

```
        if char.isupper():
```

```
            result = result + "_" + char.lower()
```

```
        else:
```

```
            result = result + char
```

```
    return result
```

```
print(convert_camel_case_to_snake_case("myVariableName"))
```

```
# 7. Convert Camel Case to Pascal Case
```

```
# Input: "myVariable"
```

```
# Output: "MyVariable"
```

```
def camel_to_pascal(text):  
    return text[0].upper() + text[1:]  
  
print(camel_to_pascal("myVariable"))
```

# Problem: Convert a string from camelCase to PascalCase.

# Input: "myVariable"

# Output: "MyVariable"

```
def convert_camel_case_to_pascal_case(text):  
    return text[0].upper() + text[1:]  
  
print(convert_camel_case_to_pascal_case("myVariable"))
```

# 8. Convert Pascal Case to Camel Case

# Input: "MyVariable"

# Output: "myVariable"

```
def pascal_to_camel(text):  
    return text[0].lower() + text[1:]  
  
print(pascal_to_camel("MyVariable"))
```

# Problem: Convert a string from PascalCase to camelCase.

# Input: "MyVariable"

# Output: "myVariable"

```
def convert_pascal_case_to_camel_case(text):
```

```
    return text[0].lower() + text[1:]
```

```
print(convert_pascal_case_to_camel_case("MyVariable"))
```

# 9. Convert Pascal Case to Snake Case

# Input: "MyVariable"

# Output: "my\_variable"

```
def pascal_to_snake(text):
```

```
    result = ""
```

```
    for char in text:
```

```
        if char.isupper() and result != "":
```

```
            result = result + "_"
```

```
            result = result + char.lower()
```

```
    return result
```

```
print(pascal_to_snake("MyVariable"))
```

# Problem: Convert a string from PascalCase to snake\_case.

# Input: "MyVariable"

# Output: "my\_variable"

```
def convert_pascal_case_to_snake_case(text):  
    result = ""  
  
    for char in text:  
        if char.isupper() and result != "":  
            result = result + "_"  
  
        result = result + char.lower()  
  
    return result  
  
print(convert_pascal_case_to_snake_case("MyVariable"))
```

# 10. Convert Text to Camel Case

# Input: "hello world example"

# Output: "helloWorldExample"

```
def text_to_camel(text):  
    words = text.split()  
    result = words[0]  
  
    for word in words[1:]:  
        result = result + word.capitalize()  
  
    return result
```

```
print(text_to_camel("hello world example"))
```

```
# Problem: Convert normal text into camelCase.
```

```
# Input: "hello world example"
```

```
# Output: "helloWorldExample"
```

```
def convert_text_to_camel_case(text):
```

```
    words = text.split()
```

```
    result = words[0]
```

```
    for word in words[1:]:
```

```
        result = result + word.capitalize()
```

```
    return result
```

```
print(convert_text_to_camel_case("hello world example"))
```

```
# 11. Convert Text to Snake Case
```

```
# Input: "hello world example"
```

```
# Output: "hello_world_example"
```

```
def text_to_snake(text):
```

```
    return "_".join(text.split())
```

```
print(text_to_snake("hello world example"))
```

# Problem: Convert normal text into snake\_case.

# Input: "hello world example"

# Output: "hello\_world\_example"

```
def convert_text_to_snake_case(text):
```

```
    return "_".join(text.split())
```

```
print(convert_text_to_snake_case("hello world example"))
```

# 12. Convert Text to Pascal Case

# Input: "hello world example"

# Output: "HelloWorldExample"

```
def text_to_pascal(text):
```

```
    words = text.split()
```

```
    result = ""
```

```
    for word in words:
```

```
        result = result + word.capitalize()
```

```
    return result
```

```
print(text_to_pascal("hello world example"))
```

# Problem: Convert normal text into PascalCase.

# Input: "hello world example"

# Output: "HelloWorldExample"

```
def convert_text_to_pascal_case(text):
```

```
    words = text.split()
```

```
    result = ""
```

```
    for word in words:
```

```
        result = result + word.capitalize()
```

```
    return result
```

```
print(convert_text_to_pascal_case("hello world example"))
```

# 13. Swap Upper and Lower Case

# Input: "HeLLo"

# Output: "hElLo"

```
def swap_upper_lower(text):
```

```
    return text.swapcase()
```

```
print(swap_upper_lower("HeLLo"))
```

# 14. Separate Digits from Text

# Input: "abc123d4"

# Output: "1234"

```
def separate_digits(text):  
    result = ""  
  
    for char in text:  
        if char.isdigit():  
            result = result + char  
  
    return result  
  
print(separate_digits("abc123d4"))
```

# 15. Print Uppercase, Lowercase, Digits,  
# and Special Characters Separately

```
def print_character_types(text):  
    uppercase = ""  
    lowercase = ""  
    digits = ""  
    special = ""  
  
    for char in text:  
        if char.isupper():  
            uppercase = uppercase + char  
        elif char.islower():  
            lowercase = lowercase + char
```

```
elif char.isdigit():  
    digits = digits + char  
else:  
    special = special + char
```

```
print("Uppercase:", uppercase)  
print("Lowercase:", lowercase)  
print("Digits:", digits)  
print("Special Characters:", special)
```

```
print_character_types("Abc123!@#")
```

# 16. Count Uppercase, Lowercase, Digits,  
# and Special Characters

```
def count_character_types(text):  
    uppercase = 0  
    lowercase = 0  
    digits = 0  
    special = 0  
  
    for char in text:  
        if char.isupper():  
            uppercase = uppercase + 1  
        elif char.islower():  
            lowercase = lowercase + 1
```

```
elif char.isdigit():  
    digits = digits + 1  
else:  
    special = special + 1
```

```
print("Uppercase:", uppercase)  
print("Lowercase:", lowercase)  
print("Digits:", digits)  
print("Special Characters:", special)
```

```
count_character_types("AbC@123x!")
```

# 17. Check Password Strength

# Input: "Pass123!"

# Output: "Strong Password"

```
def check_password_strength(password):
```

```
    uppercase = False
```

```
    lowercase = False
```

```
    digit = False
```

```
    special = False
```

```
for char in password:
```

```
    if char.isupper():
```

```
        uppercase = True
```

```
    elif char.islower():
```

```
        lowercase = True

    elif char.isdigit():
        digit = True

    else:
        special = True

    if uppercase and lowercase and digit and special:
        return "Strong Password"
    else:
        return "Weak Password"

print(check_password_strength("Pass123!"))
```

# 18. Remove Duplicates

# Input: "aabbcc"

# Output: "abc"

```
def remove_duplicates(text):
    result = ""

    for char in text:
        if char not in result:
            result = result + char

    return result
```

```
print(remove_duplicates("aabbcc"))
```

# 19. Print Duplicates

# Input: "aabbccde"

# Output: "abc"

```
def print_duplicates(text):
```

```
    result = ""
```

```
    for char in text:
```

```
        if text.count(char) > 1 and char not in result:
```

```
            result = result + char
```

```
    print(result)
```

```
print_duplicates("aabbccde")
```

# 20. Print Next Characters

# Input: "abc"

# Output: "bcd"

```
def print_next_characters(text):
```

```
    result = ""
```

```
for char in text:
```

```
    result = result + chr(ord(char) + 1)
```

```
print(result)
```

```
print_next_characters("abc")
```